Lindblad "master" equation 26/8/2024

↳ evolution of a distribution in phase space

- Generalise the Schrödinger & von Neumann unitary evolution
- Lindblad M.E. ⇒ non unitary evolution

Time	Closed	Open
discrete evolution maps	Unitary $\lvert\psi\rangle \to \lvert\psi'\rangle = U\lvert\psi\rangle$, $U^\dagger U = \mathbb{1}$ $\rho \to \rho' = U \rho U^\dagger$	Operator sum rep. $\rho_S \to \rho_S' = \sum_k E_k \rho_S E_k^\dagger$ with $\sum_k E_k^\dagger E_k = \mathbb{1}$
	↑ "generates"	↑ ?
Continuous	S.E. $\lvert\dot\psi\rangle = -iH\lvert\psi\rangle$ von Neumann $\dot\rho_S = -i[H, \rho_S]$ since $\rho_S = \lvert\psi\rangle\langle\psi\rvert$	Partial answer: $\dot\rho_{SE} = -i[H_{SE}, \rho_{SE}]$ & $\rho_S = \mathrm{tr}_E(\rho_{SE})$ $\Leftrightarrow$ $\rho_{SE}(t) = U_{SE}(t)\,\rho_{SE}(0)\,U_{SE}^\dagger(t)$

We want

$\dot\rho_S = ?$ where $\rho_S(t) = \mathrm{tr}_E(\rho_{SE}(t))$

$= -i[H_S, \rho_S] + ??$

effect of environment

$$\dot{\rho} = -i[H_S, \rho] + \sum_j \gamma_j \mathcal{D}[A_j]\rho$$

is a general form for generally open systems dynamics

The effect of the environment on the system evolution can sometimes be computed on average: Lindblad master Equation.

To derive the term(s) ?? we would need: • Stochastic Calculus

• Brownian motion

→ Financial derivatives (Black-Scholes eqn)

We are not going to use this

— we will use a "poor man's" version.

→ we will simplified analysis for a special simple case of Lindblad evolution.

For open-system evolution we start with the separability + global or unitarity assumptions:

$\Rightarrow$ $\rho_{SE} \in \mathcal{H}_S \otimes \mathcal{H}_E$ $\quad \rho_{SE}(t=0) = \rho_S(0) \otimes \rho_E(0)$

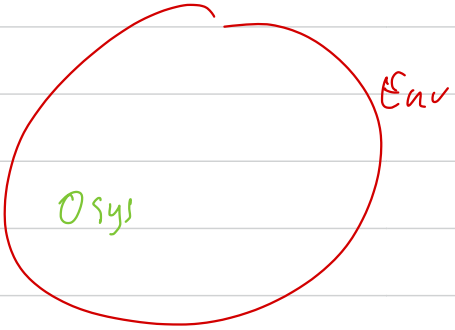
Separability:

$\mathcal{H}_S$: small dimensional; $\mathcal{H}_E$: large dimensional

global unitarity: von Neumann eqn on SOE

$$\dot{\rho}_{SE} = -i[H_{SE}, \rho_{SE}]$$

(system + env. is (approx) closed)



Reduced system density matrix $\rho_S(t)$ the $\rho_{SE}(t)$

$\Rightarrow$ $\dot{\rho}_S(t) = \frac{d}{dt} tr_{env}(\rho_{SE}) = tr_{env}\left(\frac{d\rho_{SE}}{dt}\right)$

$$= tr_{env}\left(-i[H_{SE}, \rho_{SE}(t)]\right)$$

- $C(t)$ only refers to sys. degrees of freedom
- $R(t)$ still depends on the environment

Goal: make enough simplifying approximations

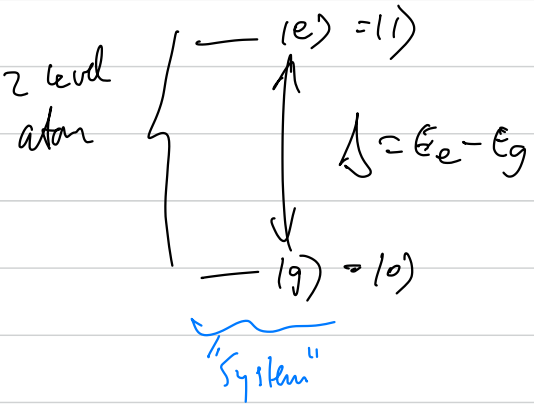
so that $R(t)$ is index of env.

$\Rightarrow$ Lindblad equation (and other variants)

We will derive a limited special case

28/8/2026

28/8/2026 Lindblad Master Equation



$$\hat{H}_{atom} = E_e |e\rangle\langle e| + E_g |g\rangle\langle g|$$

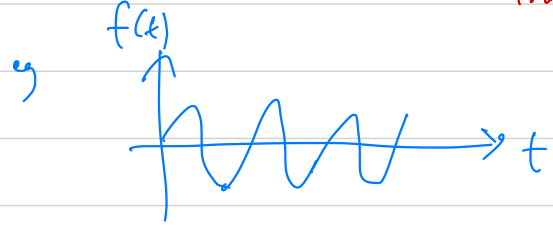
in a basis $\{|g\rangle, |e\rangle\}$

$$H_{atom} = \begin{bmatrix} E_g & 0 \\ 0 & E_e \end{bmatrix} - \frac{(E_e+E_g)}{2} \mathbb{1}$$

$$= \frac{\Delta}{2}\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \Delta = E_e - E_g$$

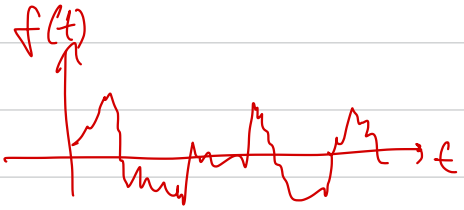
$$= \frac{\Delta}{2}\hat{z} \qquad \hat{z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- Suppose that the energy splitting Δ was time dependent: $\Delta \to \Delta(t) = \Delta_0 + \delta f(t)$
  - Δ_0: time-average value
  - $f(t)$: dimensionless function of time



eg: a known function of time

or



: a "stochastic" function of time

where $f(t)$ is a random variable

- $f(t)$ has zero mean $\langle f(t) \rangle = 0$
- randomly fluctuates

* Solve this time dependent problem

reference case is $\delta = 0$

what is the time dependent state of the system? $|\psi(t=0)\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$

$$|\psi(t)\rangle = U(t)|\psi(t=0)\rangle$$

$$= e^{-i\hat{H}_0 t}|\psi(0)\rangle$$

$$\hat{H}_0 = \begin{pmatrix}-\Delta_0/2 & 0\\ 0 & \Delta_0/2\end{pmatrix}$$

$$\Rightarrow \quad |\psi(t)\rangle = \begin{pmatrix}e^{i\Delta_0 t/2} & 0\\ 0 & e^{-i\Delta_0 t/2}\end{pmatrix}\begin{pmatrix}\alpha(0)\\ \beta(1)\end{pmatrix}$$

$$= \alpha e^{i\Delta_0 t/2}|0\rangle + \beta e^{-i\Delta_0 t/2}|1\rangle$$

$$P_g = |\langle 0|\psi(t)\rangle|^2 = |\alpha e^{+i\Delta_0 t/2}|^2 = |\alpha|^2 \quad \text{indep of time}$$

$$P_e = |\beta|^2 \quad "$$

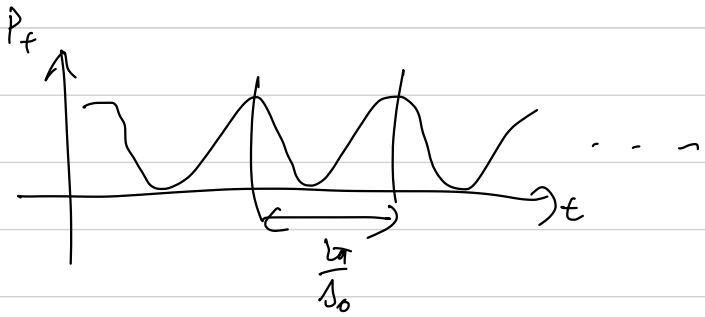
$$P_+ = |\langle +|\psi(t)\rangle|^2 = \tfrac{1}{2}\left|\alpha e^{i\Delta_0 t/2} + \beta e^{-i\Delta_0 t/2}\right|^2 \qquad |\pm\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

$$= \tfrac{1}{2}\left(1 + \alpha\beta\cos(\Delta_0 t) + \alpha\beta\sin(\Delta_0 t)\right)$$

eg $\alpha = \beta = \frac{1}{\sqrt{2}}$

$$= \tfrac{1}{2}\cdot\left(\frac{2}{\sqrt{2}}\left(\frac{e^{i\Delta_0 t/2} + e^{-i\Delta_0 t/2}}{2}\right)\right)^2 = \tfrac{1}{2}\left(\sqrt{2}\cos\Delta_0 t/2\right)^2$$

$$= \cos^2\frac{\Delta_0 t}{2} = \frac{1}{2} + \frac{1}{2}\cos\Delta_0 t$$



closed, deterministic time-indep Hamiltonian ⇒ pure evolution

"coherences" oscillate forever

- Now let's add some noise

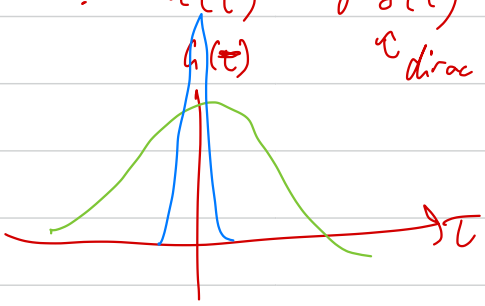
$$\hat{H}_s(t) = -\left(\Delta_0 + \underbrace{\delta f(t)}_{\xi}\right)\hat{z}$$

where



Assumptions about $f(t)$:

- zero mean at each time: $\mathbb{E}(f(t)) = 0$ (expectation)
- "stationary" noise: $\mathbb{E}[f(t)\,f(t+\tau)] = G(\tau)$ (τ correlation function)
- noise is "white noise": $G(\tau) = \gamma\,\delta(\tau)$ (δ dirac delta function)
  - "Markov" approximation



Defining property of δ-function

$$\int_{-\infty}^{\infty} dt'\, h(t')\,\delta(t') = h(0)$$

$$\Rightarrow \int_{-\infty}^{t>0} dt'\, h(t')\,\delta(t') = h(0)$$

$$+ \int_{-\infty}^{t<0} dt'\, h(t')\,\delta(t') = 0$$

$$+ \int_{-0}^{0} dt'\, h(t')\,\delta(t') \overset{?}{=} \frac{h(0)}{2}$$

Time-evolution of the state of the system

- a "closed" quantum system $\Rightarrow \dot{\rho} = -i[H(t), \rho(t)]$
- but with random stochastic noise ("classical" env.)
  $\Rightarrow$ need to average over the unobserved classical noise in $f(t)$.

$$E_f(\dot{\rho}) = E_f\left(-i\left[\hat{H}(t), \rho(t)\right]\right)$$

$$= -i\, E\left(\left[\hat{H}_0 + f(t)\,\hat{H}_1, \rho(t)\right]\right)$$

$$= -i\, E\left(\left[\hat{H}_0, \rho\right]\right) - i\, E\left(\left[f(t)\,\hat{H}_1, \rho\right]\right)$$

$$= -i\, E\left(\left[\hat{H}_0, \rho\right]\right) - i\, E\left(f(t)\left[\hat{H}_1, \rho\right]\right)$$

$$H(t) = H_0 + f(t)\,H_1$$

$$= -\frac{\Delta_0}{2}\hat{z} - f(t)\,\frac{\delta}{2}\hat{z}$$

H_0 $\quad$ $f(t)\,H_1$

evaluate $\quad E\left(\left[\hat{H}_0, \rho\right]\right) = E\left(\hat{H}_0\rho - \rho\hat{H}_0\right) = E_f(\hat{H}_0\rho) - E_f(\rho\hat{H}_0)$

$$= H_0\, E_f(\rho) - E_f(\rho)\, H_0$$

$$\mathbb{E}_f(\dot{\rho}) = -i\left[H_0, \mathbb{E}_f(\rho)\right]$$

Evaluate $\mathbb{E}_f\left(f(t)\left[H_1, \rho\right]\right) = \mathbb{E}_f\left(f(t)\, \hat{H}_1 \rho - f(t)\, \rho \hat{H}_1\right)$

$$= \hat{H}_1 \cdot \underbrace{\mathbb{E}_f\left(f(t)\rho(t)\right)}_{=0} - \underbrace{\mathbb{E}_f\left(f(t)\,\rho(t)\right)}_{=0} \cdot \hat{H}_1$$

$\rho(t)$ depends on the value of $f(t')$ for $t' < t$

claim: but $\rho(t)$ is statistically indep for $f(t')$ for $t' \geq t$

for $t' > t$ this is just "causality"

$$\mathbb{E}_f\left(f(t)\,\rho(t)\right) \overset{\text{indep.}}{=} \mathbb{E}_f\left(f(t)\right)\, \mathbb{E}_f\left(\rho(t)\right)$$

$= 0$ by zero-mean assumption

So $-i\,\mathbb{E}\left(f(t)\left[\hat{H}_1, \rho\right]\right) = 0$!

so apparently $\mathbb{E}_f(\dot{\rho}) = -i\left[H_0, \mathbb{E}_f(\rho)\right]$ ✗

$\Rightarrow \quad \dot{\bar{\rho}} = -i\left[H_0, \bar{\rho}\right]$ looks the same as evolution without noise

where $\bar{\rho} = \mathbb{E}_f(\rho)$

it's just von Neumann eqn for $\bar{\rho}$

Apparently the noise had no effect on the evolution.

Problem in this derivation

- Let's start again:

$$\dot{\rho}(t) = -i\left[\hat{H}_0 + f(t)\hat{H}_1, \rho(t)\right] \quad (\times)$$

Integrate both sides $\Rightarrow$

$$\rho(t) - \rho(t_0) = -i\int_{t_0}^{t} dt' \left[\hat{H}_0 + f(t')\hat{H}_1, \rho(t')\right] \quad (\square)$$

$\rho(t_0)$: initial state "known" independ of $f(t)$

Substitute $(\square)$ into $(\times)$

$$\Rightarrow \dot{\rho}(t) = -i\left[\hat{H}_0 + f(t)\hat{H}_1, \rho(t_0) - i\int_{t_0}^{t} dt' \left[\hat{H}_0 + f(t')\hat{H}_1, \rho(t')\right]\right]$$

$$= -i\left[\hat{H}_0, \rho(t_0)\right] - if(t)\left[\hat{H}_1, \rho(t_0)\right] - \int_{t_0}^{t} dt' \left[H_0, \left[H_0, \rho(t')\right]\right]$$

$$- \int_{t_0}^{t} \left[f(t)H_1, \left[f(t')\hat{H}_1, \rho(t')\right]\right] - \int_{t_0}^{t} dt' \left[\hat{H}_0, \left[f(t')\hat{H}_1, \rho(t')\right]\right]$$

$$- \int_{t_0}^{t} dt' \left[f(t)\hat{H}_1, \left[\hat{H}_0, \rho(t')\right]\right]$$

As before, we should take Expectations of LHS + RHS over the random function $f(t)$

$$E_f(\text{"}\text{"}) =$$

$$\dot{\rho} = -i\left[\hat{H}_0, \rho(t_0)\right] - i f(t)\left[\hat{H}_1, \rho(t_0)\right] - \int_{t_0}^{t} dt' \left[H_0, \left[H_0, \rho(t')\right]\right]$$

$$- \int_{t_0}^{t} \left[f(t) H_1, \left[f(t') \hat{H}_1, \rho(t')\right]\right] - \int_{t_0}^{t} dt' \left[\hat{H}_0, \left[f(t') \hat{H}_1, \rho(t')\right]\right]$$

$$- \int_{t_0}^{t} dt' \left[f(t) \hat{H}_1, \left[\hat{H}_0, \rho(t')\right]\right]$$

$E_f(f(t)) = 0$

so $E_f(\dot{\rho}) = \dot{\bar{\rho}}$

$$= -i\left[\hat{H}_0, \bar{\rho}(t_0)\right] + 0 - \int_{t_0}^{t} dt' \left[H_0, \left[H_0, \bar{\rho}(t')\right]\right]$$

$$- E_f\left(\int_{t_0}^{t} dt' f(t) f(t') \left[H_1, \left[H_1, \rho(t')\right]\right]\right)_{①} - E_f\left(\int_{t_0}^{t} dt' f(t') \left[H_0, \left[H_1, \rho(t')\right]\right]\right)_{②} \quad (② = 0)$$

$$- E_f\left(\int_{t_0}^{t} dt' f(t) \left[H_1, \left[H_0, \rho(t')\right]\right]\right)_{③} \quad (③ = 0)$$

① : $t > t'$ & $f(t)$ & $f(t')$ are statistically uncorrelated for $t \neq t'$

& $\rho(t')$ is statistically uncorrelated with $f(t)$, $t \geq t'$

so $E_f\left(f(t) f(t') \rho(t')\right) = \underbrace{E_f\left(f(t) f(t')\right)}_{= G(t-t') = \gamma \delta(t-t') \text{ by assumption}} \underbrace{E_f(\rho(t'))}_{= \bar{\rho}}$

$$① = -\int_{t_0}^{t} dt' E_f\left(f(t) f(t')\right) \left[H_1, \left[H_1, \bar{\rho}(t')\right]\right]$$

$$= -\int_{t_0}^{t} dt' \gamma \delta(t-t') \left[H_1, \left[H_1, \bar{\rho}(t')\right]\right]$$

$$= -\frac{\gamma}{2} \left[H_1, \left[H_1, \bar{\rho}(t)\right]\right]$$

② $= -\int_{t_0}^{t} dt' E_f\left(f(t')\right) \left[H_0, \left[H_1, \bar{\rho}(t')\right]\right]$ (since $E_f\left(f(t) \rho(t')\right) = E_f\left(f(t')\right) E_f\left(\rho(t')\right)$, uncorrelated R.V)

$= 0$ ($E_f(f(t')) = 0$)

③ $= 0$ too since $E_f\left(f(t) \rho(t')\right) = E_f\left(f(t)\right) \bar{\rho}$ ($E_f(f(t)) = 0$; $f(t)$, $\rho(t')$ correlated R.V.s)

$\Rightarrow$

$$\dot{\bar{\rho}} \approx -i[H_0, \bar{\rho}(t_0)] - \int_{t_0}^{t} dt' [H_0, [H_0, \bar{\rho}(t')]]$$

$$- \frac{\gamma}{2}[H_1, [H_1, \tilde{\rho}(t)]]$$ ← the effect of fluctuations in $f(t)$

$$-i[H_0, \bar{\rho}(t_0)] - \int_{t_0}^{t} dt' [H_0, [H_0, \bar{\rho}(t')]] = -i\left[H_0, \bar{\rho}(t_0) - i\int_{t_0}^{t} dt' [H_0, \tilde{\rho}(t')]\right] \quad (*)$$

claim: $\dot{\bar{\rho}} \approx -i[H_0, \bar{\rho}(t)]$

$\Rightarrow$ $\boxed{\dot{\tilde{\rho}} = -i[H_0, \tilde{\rho}] - \frac{\gamma}{2}[H_1, [H_1, \tilde{\rho}]]}$ ← Lindblad Master equation.

"proof of claim" if $\dot{\bar{\rho}} = -i[H_0, \bar{\rho}] + O(\gamma)$

$$\Rightarrow \bar{\rho}(t) - \bar{\rho}(t_0) = -i\int_{t_0}^{t} dt' [H_0, \bar{\rho}(t')] + \int_{t_0}^{t} O(\gamma)$$

$$\Rightarrow \bar{\rho}(t) = \bar{\rho}(t_0) - i\int_{t_0}^{t} dt' [H_0, \bar{\rho}(t')] + O(\gamma)$$

So $(*) = -i[H_0, \tilde{\rho}(t)] + O(\gamma)$

So $\boxed{}$ is self-consistent.

So the evolution of the system is governed by the Lindblad master Equation

$$\dot{\rho} = -i[H_0, \rho] + \gamma \mathcal{D}[H_1]\rho$$

$$\{ -\tfrac{\gamma}{2}[H_1,[H_1,\rho]] \} = -\gamma\left(\left(\tfrac{H_1H_1\rho}{2} - H_1\rho H_1\right) - \left(\tfrac{H_1\rho H_1}{2} - \rho H_1 H_1\right)\right)$$

$$= \gamma\left(H_1\rho H_1 - \tfrac{1}{2}(H_1H_1\rho + \rho H_1 H_1)\right)$$

Check $\Rightarrow \mathcal{D}[H_1]\rho = H_1\rho H_1^\dagger - \tfrac{1}{2}\left(H_1^\dagger H_1\rho + \rho H_1^\dagger H_1\right)$

$= H_1\rho H_1 - \tfrac{1}{2}(H_1H_1\rho + \rho H_1 H_1)$ if $H_1 = H_1^\dagger$

where $\mathcal{D}[A]\rho = A\rho A^\dagger - \tfrac{1}{2}(A^\dagger A\rho + \rho A^\dagger A)$

"dissipative superoperator"

Same class of mathematical object that act on state matrices.

$[H,\rho]$ is a "coherent superoperator"

Operators ($n \times 1$ matrices) act on n-dim state vectors $|\psi\rangle$

Superoperators ($n^2 \times n^2$ matrices) " " n^2-dim state operators ρ

Think of system density operators as n^2-dim vectors

+ Superoperators are a matrix representation of commutators & dissipators.

Dimensions

$$\left(|\dot{\psi}\rangle = -i \frac{H}{\hbar} |\psi\rangle \right) \qquad [\hbar] = J_s \qquad [H] = J$$

$$\tilde{H}_s(t) = -\left(\Delta_0 + \underbrace{f(t)}_{x}\right)\hat{z} = H_0 + H_1 f(t)$$

$$H_0 = -\frac{\Delta_0}{2}\hat{z}, \quad H_1 = -\frac{1}{2}\hat{z}$$

$$[\Delta_0] = J \Rightarrow [H_0] = J \quad + \quad [H_1] = 1$$

$$\dot{\rho} = -i\left[\frac{H_0}{\hbar}, \rho\right] + \frac{1}{\hbar^2}\frac{\gamma}{2}\mathcal{D}[H_1]\rho \qquad (*)$$

where $E_f(f(t)f(t')) = \gamma\,\delta(t-t')$ (D)

$$\int_{-\infty}^{\infty} dx\, \delta(x) = 1$$

from (*) $\left[\frac{\gamma}{\hbar^2}\right] = s^{-1} \Rightarrow [\gamma] = J^2 s$

So from (D) $[RHS] = [J^2 s][\delta(t)] = [J^2 s][s^{-1}] = [J^2]$

from LHS (D) $[LHS] = [f^2] = [J^2]$

$\Rightarrow [f] = [J]$

& so $\left[\frac{\gamma}{\hbar^2}\right] = s^{-1}$ is a rate

For one two level atom driven by white noise, the evolution is given by

$$\dot{\rho} = -i\left[\Delta_0 \hat{z}, \rho\right] + \gamma \mathcal{D}\left[\hat{z}\right]\rho$$

accounts for the noise acting on the system

where $\gamma = \gamma/t^2$

(& $\rho = \bar{\rho}_s$ is the noise-averaged state of the system)

This is a general form that applies to many systems that are weakly coupled to an environment.

Representations of linear transformations on a vector space $\mathbb{C}^n$

$|\psi\rangle \in \mathbb{C}^n$: $|\psi\rangle \to |\psi'\rangle = \underset{\sim}{A}\,\underset{\sim}{|\psi\rangle}$

$[\ ]_{n\times n}\ [\ ]_{n\times 1}$

$\langle\psi| \to \langle\psi'| = \underset{\sim}{\langle\psi'|}\,A^\dagger$

$[\ ]_{n\times 1}\ (\ )^\dagger_{n\times n}$

$\rho \in \mathbb{C}^n \times \mathbb{C}^n$ (matrix)

$\rho \to \rho' = [H,\rho] = H\rho - \rho H$

$\rho \to \rho' = \mathcal{D}[A]\rho = A\rho A^\dagger - \frac{1}{2}(A^\dagger A\rho + \rho A^\dagger A)$

or $\rho \in \mathbb{C}^{n^2}$ (Supervector)

$\rho \to \rho' = \big([H,\cdot]\big)\rho$

$= [\ ]_{n^2\times n^2}\ [\ ]_{n^2\times 1}$

or $\rho \to \rho' = \big(\mathcal{D}[A]\cdot\big)\rho$

$= [\ ]_{n^2\times n^2}\ [\ ]_{n^2\times 1}$

For numerical calculations:

$$|\dot{\psi}\rangle = -iH|\psi\rangle$$

$$\Rightarrow \quad |\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$$

matrix exponential

$$\underline{\underline{A}} = \underline{\underline{P}}\cdot\underline{\underline{D}}\cdot\underline{\underline{P}}^{\dagger} \text{ if } \underline{\underline{A}} = \underline{\underline{A}}^{\dagger} \quad \text{with} \quad \underline{\underline{P}}^{\dagger}\underline{\underline{P}} = \mathbb{1}$$

$$\Rightarrow \quad e^{\underline{\underline{A}}} = e^{\underline{\underline{P}}\cdot\underline{\underline{D}}\cdot\underline{\underline{P}}^{\dagger}} = \underline{\underline{P}}\, e^{\underline{\underline{D}}}\, \underline{\underline{P}}^{\dagger}$$

How about $\dot{\underline{x}} = \underline{\underline{A}}\,\underline{x}$ where $\underline{\underline{A}}$ is an arbitrary matrix

$$\Rightarrow \quad \underline{x}(t) = e^{\underline{\underline{A}}t}\,\underline{x}(0)$$

Then we have to revert to the definition

$$e^{\underline{\underline{A}}t} = \sum_{j=0}^{\infty} \frac{(\underline{\underline{A}}t)^j}{j!} = \sum_{j=0}^{\infty} \frac{\underline{\underline{A}}^j t^j}{j!}$$

eg $\dot{\rho} = -i[H,\rho]$ Von Neumann eqn

$$\Rightarrow \quad \rho(t) = e^{-iHt}\,\rho(0)\,e^{iHt} \quad \text{- solution.}$$

$$\dot{\rho} = -i[H,\rho] + \gamma\mathcal{D}[A]\rho \quad \text{- Lindblad M.E}$$

$$\dot{\rho} = \underline{\mathcal{B}}[\rho] \quad \text{where } \mathcal{B}[\cdot] = -i[H,\cdot] + \gamma\mathcal{D}[A]\cdot$$

"vectorise" or flatten $\rho \longrightarrow [F]_{n^2 \times 1}$

$$\dot{f}_{n^2 \times 1} = \underset{=}{B}_{n^2 \times n^2} \cdot f_{n^2 \times 1}$$

$$\Rightarrow \quad f(t) = e^{\underset{=}{B}t} f(0)$$